

1955



The first radiometric dating of rock layers, such as these in Bryce Canyon, USA, put estimates of Earth's age at only about 2 billion years, far below the actual age of the planet.

IN 1955, AMERICAN GEOCHEMIST CLAIR PATTERSON (1922–95) studied the atoms of rocks to estimate **the age of Earth**. He focused his analysis on meteorites, which are considered remnants of an age when the Solar System—and therefore Earth—was first made. Patterson **isolated lead from samples of rock** and studied the relative proportions of lead isotopes—variants of the metal. **Radioactive atoms**, such as those of uranium, decay at a known rate (see p.267), and thereby provide information about the age of samples (see below). Some of these atoms decay into specific lead isotopes,

4.55

BILLION YEARS THE AGE OF EARTH

which build up over time. Patterson calculated Earth's age as 4.55 billion years—older, and more accurate than prevailing estimates. This changed the way scientists viewed our world.

British physicist **Louis Essen** (1908–97), working at the UK's National Physics Laboratory, designed the **first atomic clock**. Its time-keeping was based on radiation emitted by atoms of

cesium metal. It would make **time measurement more precise** than ever—the new atomic clock would gain or lose only a second in 300 years. Modern atomic clocks are accurate to 1 second in 6 million years.

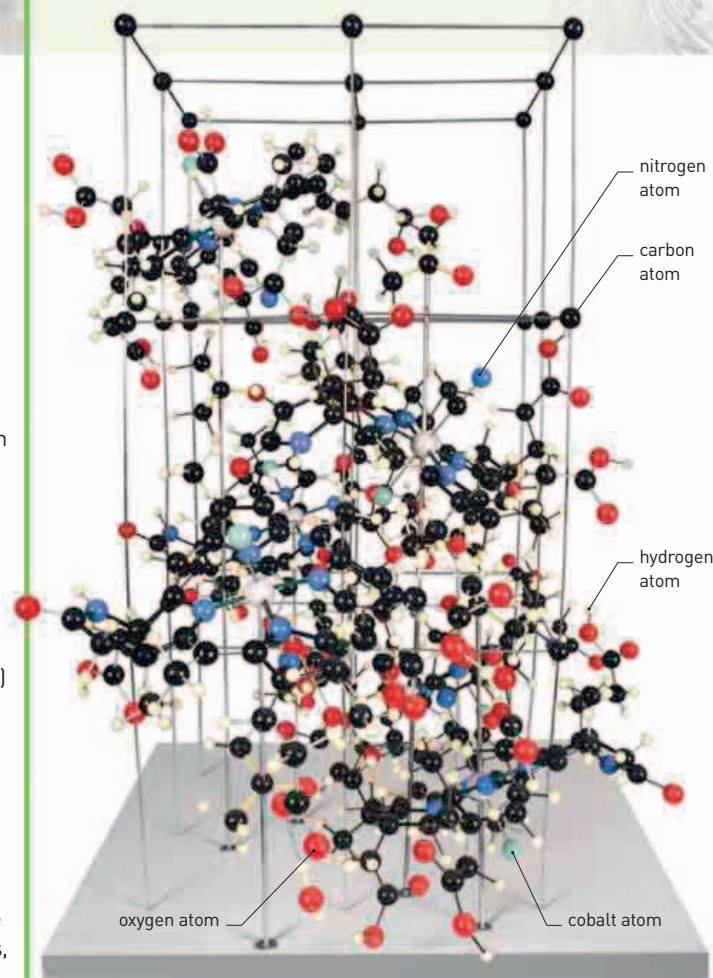
In December, in the laboratory of Swedish geneticist Albert Levan at the Institute of Genetics in Lund, Sweden, visiting Japanese-born American biologist **Joe Hin Tjio** (1919–2001) made a discovery that corrected a 50-year error in the field of **genetics**: that human cells contained 48 chromosomes. He showed that there are in fact **46 chromosomes in a normal human body cell**. Tjio used an improved microscopic technique to squash cells into single layers, rather than relying on slicing thin sections of tissue. He also used a technique to spread out the tiny chromosomes in a sample, so that they separated easily and clearly, without fragmenting. Tjio continued his career in the US, where he became a significant figure in developing the new branch of biology called **cytogenetics** (cell genetics).

1956

2,244

MILES

THE LENGTH OF **TRANSATLANTIC TELEPHONE CABLE**

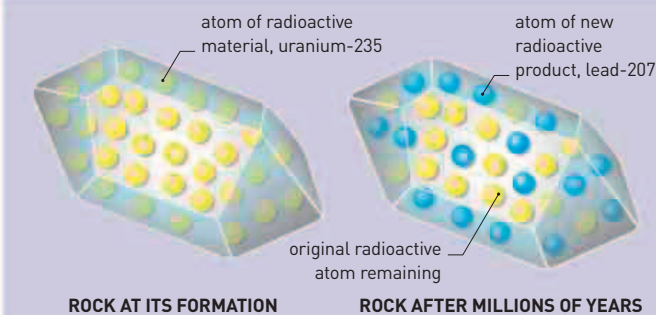


Structure of vitamin B12

A model of a vitamin B12 molecule reveals the complexity of its structure. This was first revealed by Dorothy Hodgkin, through a method called X-ray crystallography (see 1945).

metal poisoning. In the years that followed, the disease was linked specifically to mercury that had been leaking into the

A NEW DISEASE affecting a fishing community in Minamata, Japan, was described in May 1956. Called **Minamata disease**, it was characterized by progressive paralysis reminiscent of many nervous system disorders. It was not until November that the disease was traced to **heavy**



RADIOMETRIC DATING

Radiometric dating is a technique used to date rocks, minerals, or fossils using natural rates of radioactive decay. Its origins are in the work done by physicists in the early 1900s. Radioactive material decays at a known rate. By working out the ratio of a radioactive material (for example, uranium) to its decay products (lead), the age of the rock can be calculated.

American geochemist Clair Patterson uses radiometric dating to recalculate **age of Earth**

August 24 British physicist Louis Essen launches the first **workable atomic clock**

April 12 US Food and Drug Administration **approves Salk's polio vaccine**

December 22 Japanese-born biologist Joe Hin Tjio discovers that **human cells contain 46, not 48, chromosomes**

May 1 Minamata disease described

July 14 Vitamin B12 structure is published

September 25 Launch of transatlantic submarine telephone cable